17 FILE HANDLING • THEORY

01 fstream classes and modes

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — File Stream Classes (`ofstream`, `ifstream`, `fstream`) and Opening Modes



fstream Hierarchy

1. `std::ofstream`: Output file stream (for writing to files; default mode `ios::out | ios::trunc`).
2. `std::ifstream`: Input file stream (for reading from files; default mode `ios::in`).
3. `std::fstream`: Bidirectional file stream (default mode `ios::in | ios::out`).

Mode Flag	Meaning
`ios::in`	Open for reading
`ios::out`	Open for writing
`ios::app`	Append all writes to the end of the file
`ios::ate`	Move initial file position to the end upon opening
`ios::trunc`	Discard existing file contents if the file already exists
`ios::binary`	Open in binary mode (no newline character translation)